

STATEMENT OF WORK

1. Objective/Requirements

This project has the goal of building a wind lidar prototype that operates at 780.2 nm. The prototype requires two Fabry Perot etalons as part of the receiver optics subsystem.

2. Characteristics, Scope, and Specs

The contractor shall provide the following items:

Solid Etalon

- Etalon plate diameter: 60.0 mm (± 0.25 mm)
- Etalon thickness: 46.0 mm (± 0.1 mm)
- Clear (coated) aperture: 57 mm
- Surface flatness: $\lambda/70$ at 632 nm (best effort)
- Free Spectral Range: 2.24 GHz at 780.2 nm
- Parallelism: $\lambda/65$ at 632 nm (best effort)
- Effective Finesse: 16 at 780.2 nm over 55 mm aperture (best effort)
- Transmission: 81% at 780.2 nm (best effort)
- Ultra-hard high energy coatings
- Mounted in 70.0 mm diameter x 48.0 mm aluminum cell with protective dust covers
- Qty needed: 1

Air-Gapped Etalon

- Etalon plate diameter: 66.0 mm
- Etalon thickness: 15 mm
- Spacers (physical): 1.521 mm (± 0.001 mm) thick Corning ULE or Schott Zerodur
- Clear (coated) aperture: 54 mm
- Surface flatness: $\lambda/75$ at 632 nm (best effort)
- Outer surfaces wedged at 30 arcmin, matched and opposed
- Free Spectral Range: 98.5 GHz at 780.2 nm
- Parallelism: $\lambda/75$ at 632 nm (best effort)
- Effective Finesse: 20 at 780.2 nm over 50 mm aperture (best effort)
- Transmission: 84% at 780.2 nm (best effort)
- Ultra-hard high energy coatings
- Mounted in 76.2 mm diameter x 34.5 mm aluminum cell with protective dust covers
- Qty needed: 1

3. Deliver to:

Daniel Cremons

NASA GSFC, Code 698

Building 33, Room D406
8800 Greenbelt Dr.
Greenbelt, MD 20771

4. Period of Performance: 21 weeks ARO